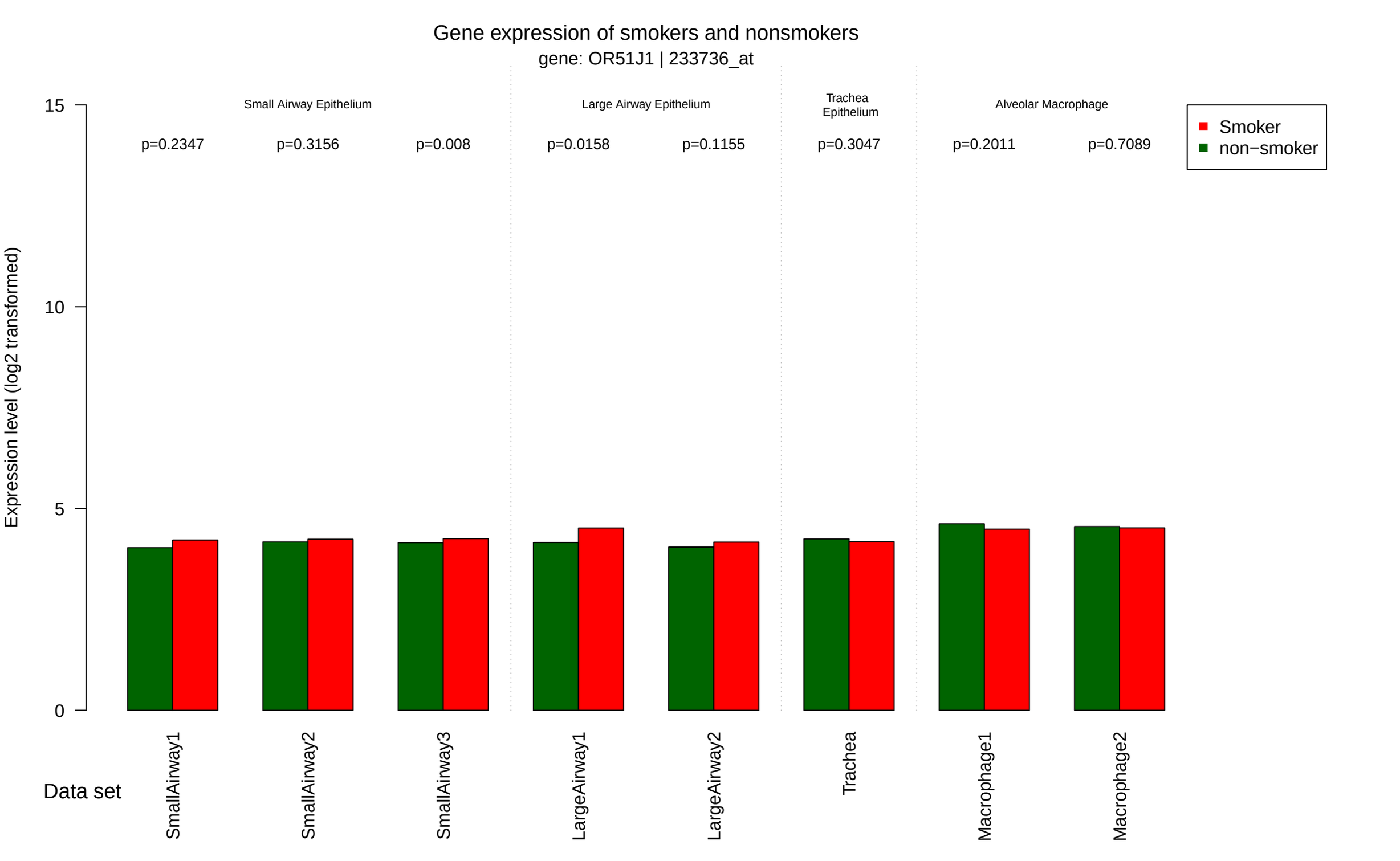
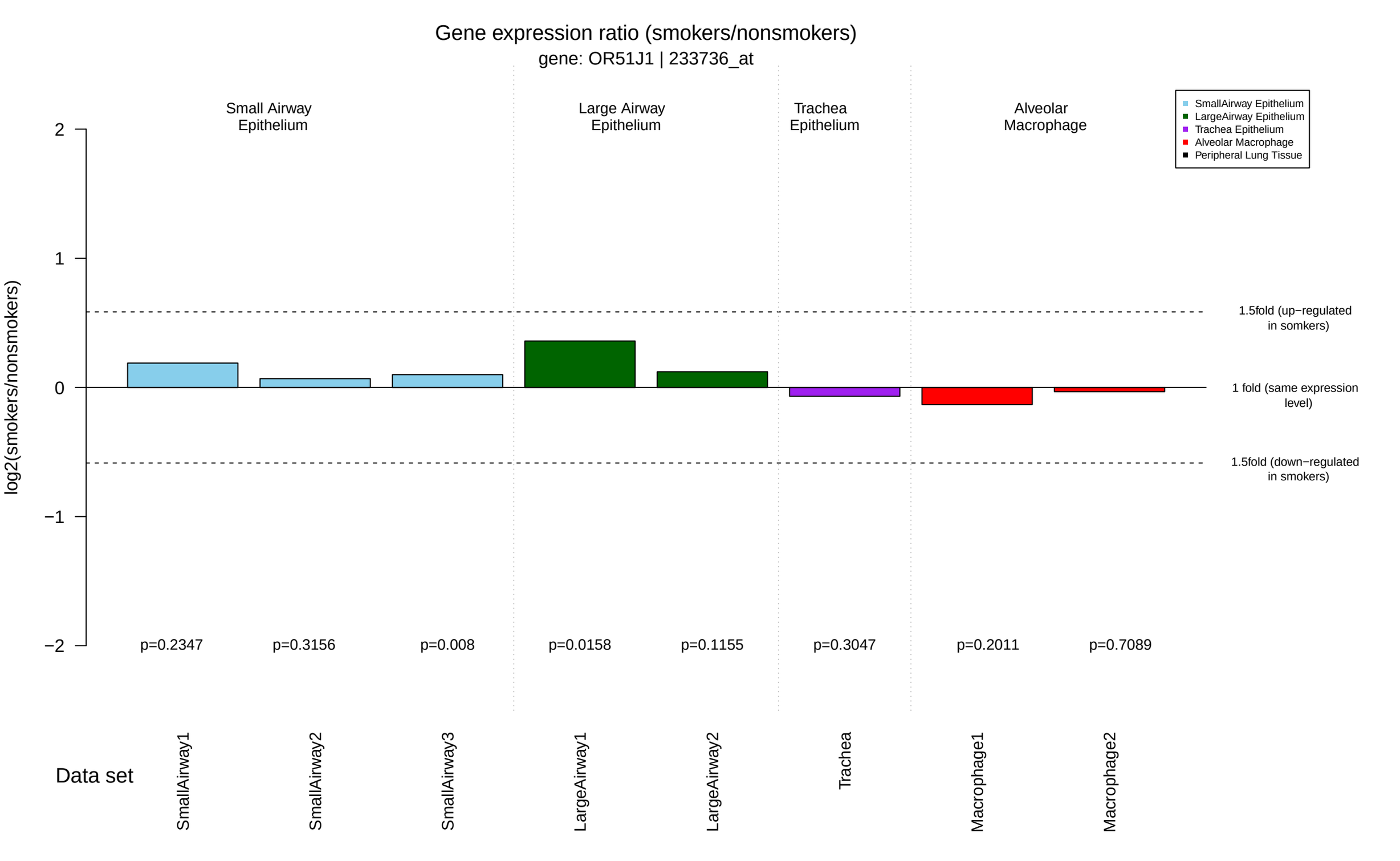
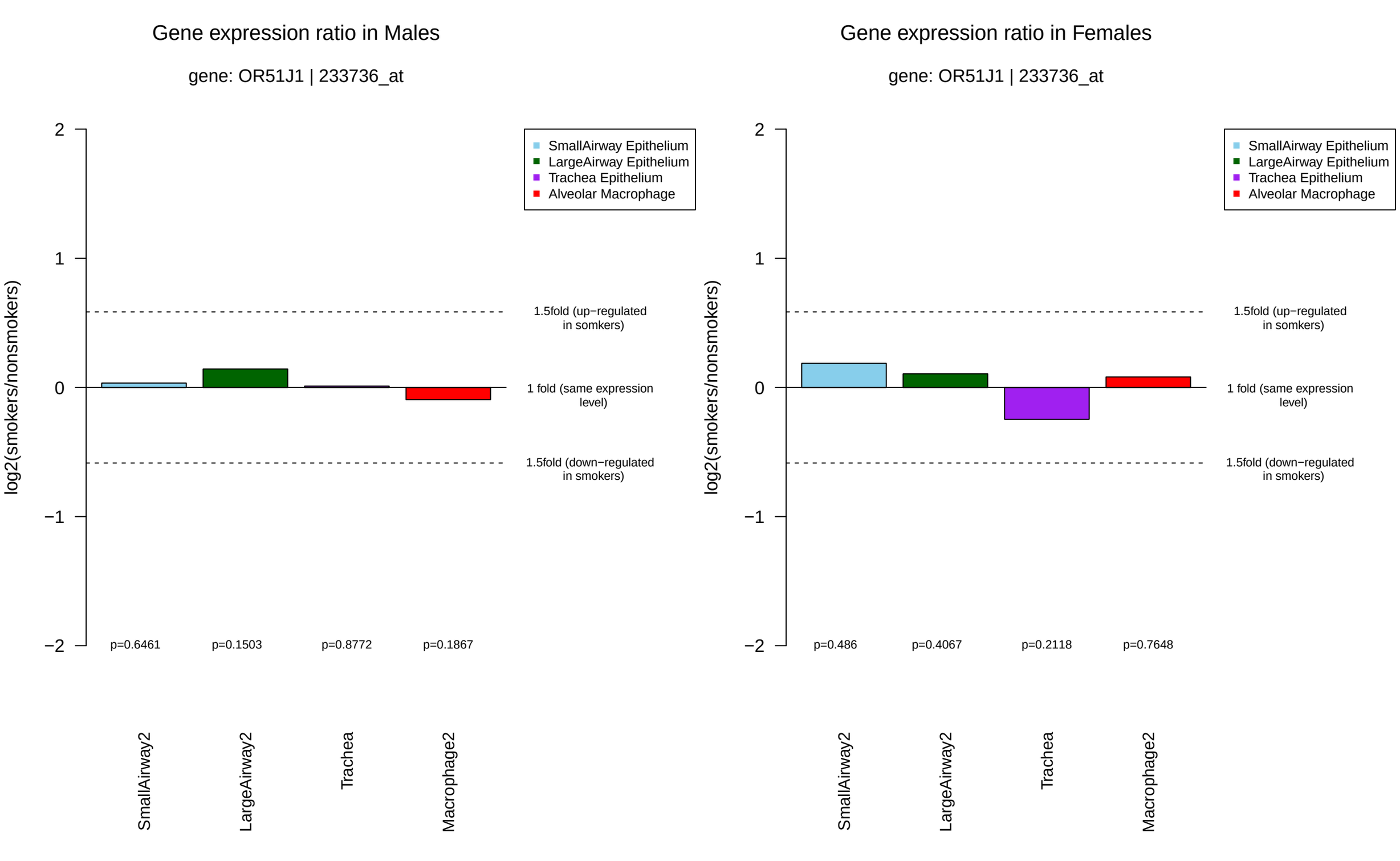
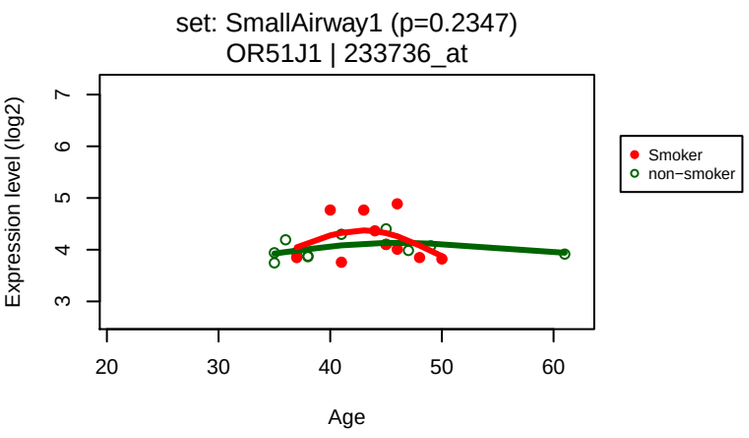
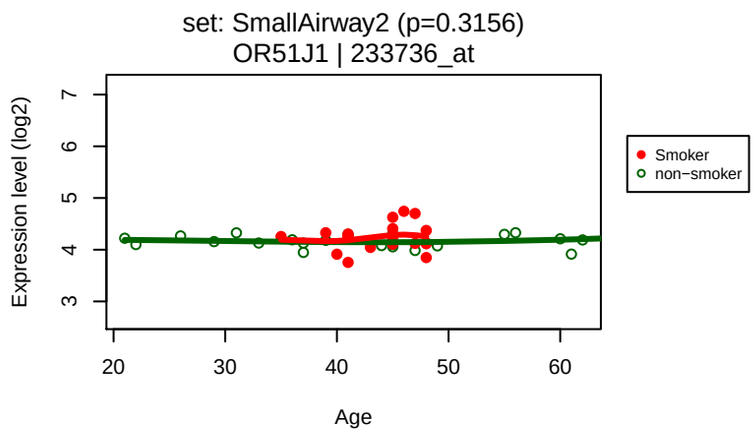
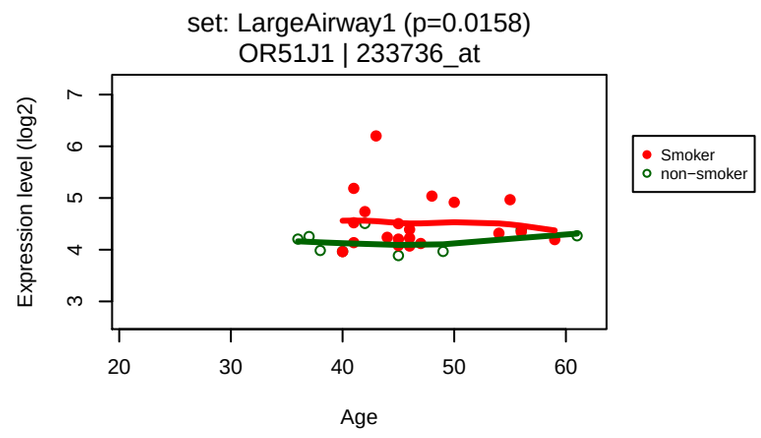

Gene expression of smokers and nonsmokers
gene: OR51J1 | 233736_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2347
p=0.3156
p=0.008
p=0.0158
p=0.1155
p=0.3047
p=0.2011
p=0.7089
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

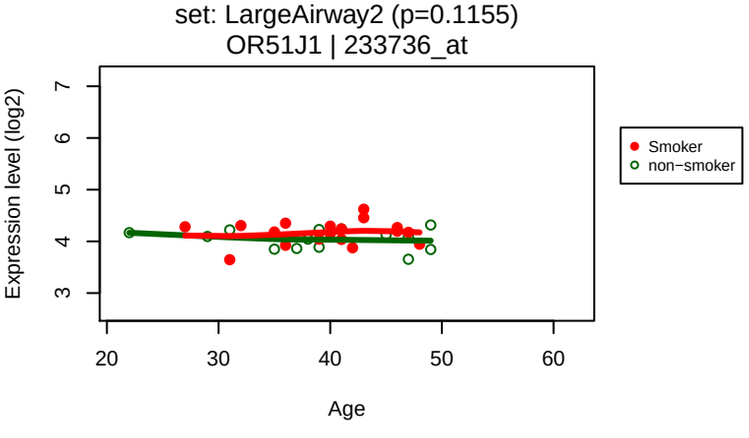

Gene expression ratio (smokers/nonsmokers)
gene: OR51J1 | 233736_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2347
p=0.3156
p=0.008
p=0.0158
p=0.1155
p=0.3047
p=0.2011
p=0.7089
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

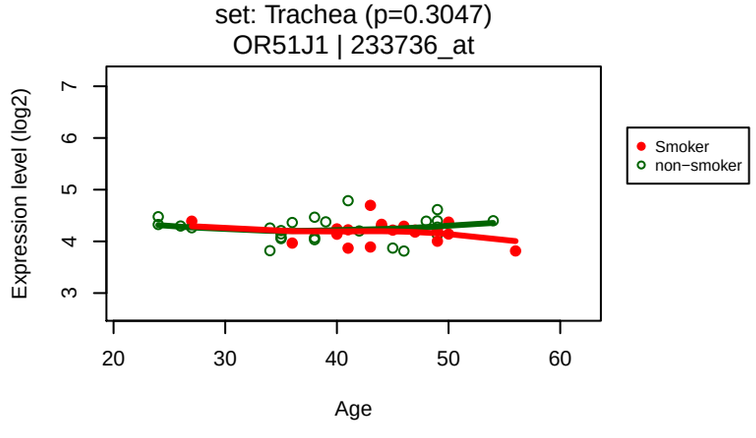

Gene expression ratio in Males
gene: OR51J1 | 233736_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6461
p=0.1503
p=0.8772
p=0.1867
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR51J1 | 233736_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.486
p=0.4067
p=0.2118
p=0.7648
SmallAirway2
LargeAirway2
Trachea
Macrophage2

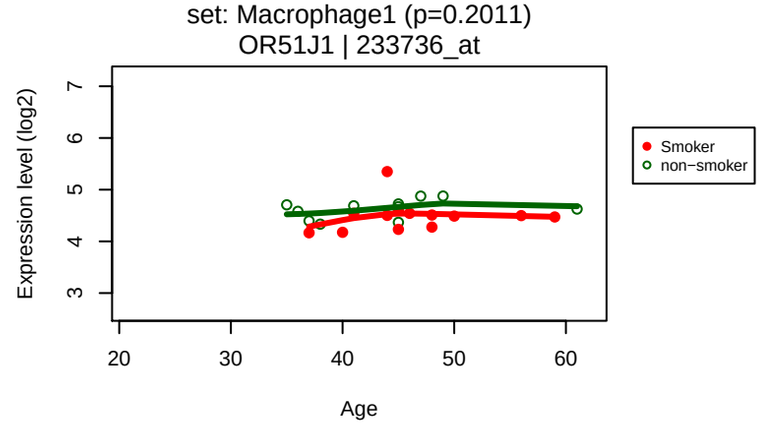

set: SmallAirway1 (p=0.2347)
OR51J1 | 233736_at
Expression level (log2)
Age
Smoker
non−smoker

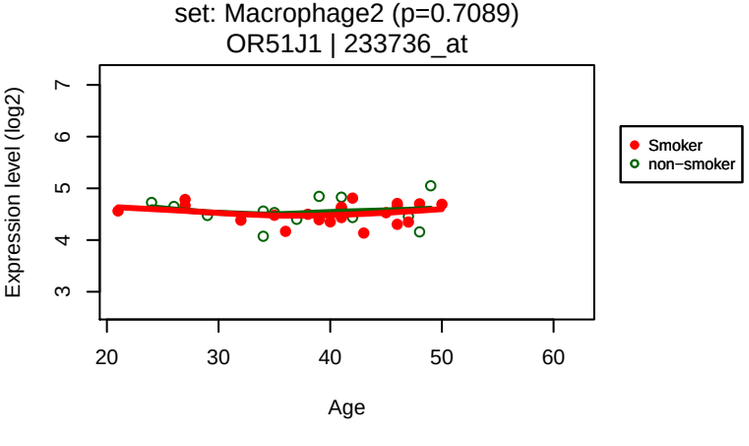

set: SmallAirway2 (p=0.3156)
OR51J1 | 233736_at
Expression level (log2)
Age
Smoker
non−smoker

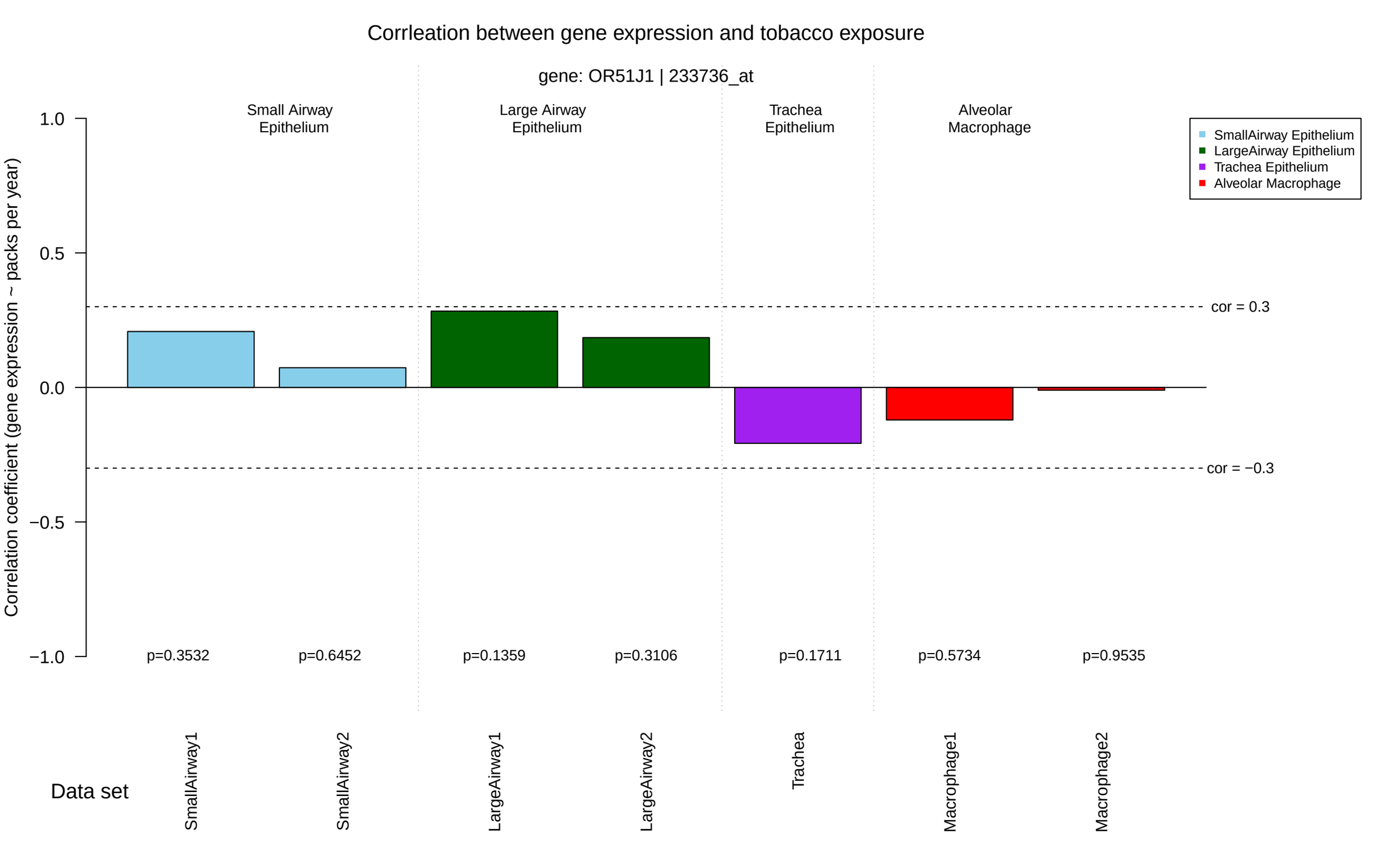

set: LargeAirway1 (p=0.0158)
OR51J1 | 233736_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1155)
OR51J1 | 233736_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3047)
OR51J1 | 233736_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2011)
OR51J1 | 233736_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7089)
OR51J1 | 233736_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR51J1 | 233736_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3532
p=0.6452
p=0.1359
p=0.3106
p=0.1711
p=0.5734
p=0.9535
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2